
MA 555: Introduction to Partial Differential Equations for Engineers

Course Code	MA 555
Course Name	Introduction to Partial Differential Equations for Engineers
L-T-P-C	3-0-0-3
Prerequisites	IC 110: Engineering Mathematics
Intended for	UG/PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Introduction** [6 Lectures]
Overview of PDEs, Classification of second order equations, Initial value problems, boundary value problems.
- **Hyperbolic Partial Differential Equations** [8 Lectures]
Introduction to the wave equation, the method of spherical means, Kirchhoff's formula and Minkowskian geometry, geometric energy estimates.
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- **Elliptic Partial Differential Equations** [8 Lectures]
Introduction to Laplace's and Poisson's equations, fundamental solution, Green functions, Poisson's formula, Harnack's inequality, Liouville's theorem.
- **Fourier transform** [6 Lectures]
Introduction to the Fourier transform; Fourier inversion and Plancherel's theorem.
- **Special Equations** [6 Lectures]
Introduction to Schrodinger's equation; Introduction to Lagrangian field theories, Transport equations and Burger's equation

References

1. Sandro Salsa, **Partial Differential Equations in Action: From Modelling to Theory**, Springer, 2010.
2. Robert C. McOwen, **Partial Differential Equations - Methods and Applications**, Pearson Education Inc., 2004.
3. S.J. Farlow, **Partial Differential Equations for Scientists and Engineers**, Dover Publications, New York, 1982.
4. E. C. Zachmanoglou and Dale W. Thoe, **Introduction to Partial Differential Equations with applications**, Dover Publications, 1988.
5. Gerald B. Folland, **Introduction to Partial Differential Equations**, Princeton University Press, 1995.